PhiCor.ai

The Mathematical Human Body

Comprehensive Feature Documentation

1,977	336+	18	10
Disease Models	Drug Profiles	Organ Systems	Biological Scales

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1. Overview

PhiCor.ai is a comprehensive digital twin platform that mathematically models the human body across multiple biological scales. It provides clinicians, researchers, and medical experts with an interactive tool to explore physiological parameters, disease modifications, drug effects, and differential diagnoses.

Key Features:

- **1,977 Disease Models** - AI-enhanced parameter modifications
- **336+ Drug Profiles** - Complete physiological effects modeling
- **18 Organ Systems** - Heart, Brain, Lungs, Liver, Kidneys, and more
- **10 Biological Scales** - Geometry, Pressures, ECG, Cellular, Lab Tests, etc.
- **Expert Curation System** - Medical accuracy validation by specialists
- **Knowledge Bank** - AI-powered continuous updates from medical literature

Technology Stack:

- **Frontend:** React.js with Tailwind CSS & shadcn/ui
- **Backend:** FastAPI (Python)
- **Database:** MongoDB
- **AI:** OpenAI GPT-4o via Emergent LLM Key

2. Main Clinical Dashboard

The main dashboard provides an interactive interface for exploring physiological parameters across all organ systems. Users can select organs from the body map, view parameters by scale, and apply disease modifications to see how values change.

Dashboard Components:

- **Body Map (Left):** Interactive visualization of 18 organ systems
- **Parameter Grid (Center):** Multi-scale parameter display with values and ranges
- **Disease Search (Top):** Search and apply disease modifications
- **Feature Panels (Right):** Quick access to Drugs (ρ), Differentials (Δ), Calculator (λ), Timeline (Ω)

3. Disease Models with Examples

PhiCor contains 1,977 disease models with AI-enhanced parameter modifications. Each disease includes detailed parameter changes across relevant organ systems and scales.

Example: Atrial Fibrillation

When Atrial Fibrillation is applied, the following parameter modifications occur:

Parameter	Normal	Modified	Change
PR Interval	160 ms	ABSENT	No P waves
Heart Rate	72 bpm	100-180 bpm	Irregular
QTc Interval	420 ms	480 ms	+14.3%
Cardiac Output	5 L/min	4.25 L/min	-15%
P Wave	Present	Absent	Fibrillatory waves

4. Drug Engine

The Drug Engine contains 336+ drug profiles with detailed physiological effects. Drugs are categorized by therapeutic class and show how they modify body parameters.

Drug Categories:

- **Cardiovascular:** ACE inhibitors, Beta-blockers, Antiarrhythmics, Diuretics
- **Respiratory:** Bronchodilators, Corticosteroids, Antihistamines
- **CNS:** Antidepressants, Antipsychotics, Anxiolytics, Anticonvulsants
- **Antimicrobial:** Antibiotics, Antivirals, Antifungals
- **Oncology:** Chemotherapeutics, Targeted therapies, Immunotherapies
- **And more...** Endocrine, GI, Hematological, Analgesics, Psychiatric

5. Differential Diagnosis

The AI-powered differential diagnosis panel analyzes abnormal parameters and generates a ranked list of possible diagnoses with probability scores and key distinguishing features.

How It Works:

1. Select an organ system with abnormal parameters
2. AI analyzes the parameter pattern
3. Generates ranked differential diagnoses
4. Each diagnosis shows: probability score, supporting findings, differentiating features

6. Expert Curation Portal

The Expert Curation Portal allows medical specialists to review, verify, and update the platform's data. Experts can suggest changes to parameters, which are then reviewed and approved by administrators.

Portal Tabs:

- **Baselines:** Review normal/standard parameters by organ and scale
- **Diseases:** Review disease-specific parameter modifications (1,581 diseases)
- **References:** Lab tests, guidelines, and protocols (396 items)
- **Changes:** Pending change requests for approval
- **Guidelines:** AI-detected guideline updates
- **Lab:** Experiment sandbox for testing
- **Knowledge:** Upload documents for AI analysis (Admin only)
- **Users:** User management (Admin only)

7. Knowledge Bank

The Knowledge Bank allows administrators to upload medical documents, URLs, and text for AI analysis. The system automatically extracts parameter updates, guideline changes, and clinical modifications, creating suggestions for admin approval.

Features:

- **Upload Sources:** URLs, PDFs, Documents, or direct text input
- **AI Analysis:** GPT-4o extracts medical updates automatically
- **Suggestions:** Review proposed changes with evidence levels and confidence scores
- **Approval Workflow:** Approve or reject each suggestion
- **Auto-Search:** Scheduled AI search for new guidelines (configurable 7-90 days)

8. User Management

The User Management system allows Super Admins to create, edit, and manage user accounts. Users are assigned roles with different permission levels.

User Roles:

Role	Permissions
Expert	View parameters, suggest changes, run experiments
Admin	All Expert + approve changes, view audit logs
Super Admin	All Admin + manage users, system settings

9. API Reference

PhiCor.ai provides a comprehensive REST API for integration with external systems. All protected endpoints require JWT authentication.

Core Endpoints:

Endpoint	Method	Description
/api/experts/login	POST	Authenticate and get JWT token
/api/experts/diseases-to-review	GET	Get all diseases with modifications
/api/disease/{id}/parameters	GET	Get parameters for specific disease
/api/change-requests	POST	Submit parameter change request
/api/admin/knowledge/upload	POST	Upload knowledge source
/api/admin/knowledge/suggestions	GET	Get pending AI suggestions
/api/admin/experts	GET/POST	List/Create experts
/api/admin/settings	GET/PUT	System settings

Authentication:

All protected endpoints require the Authorization header: **Authorization: Bearer {token}**